Frequency Response

Figure 15-2 shows the frequency response of the moving average filter. It is mathematically described by the Fourier transform of the rectangular pulse, as discussed in Chapter 11:

EQUATION 15-2
Frequency response of an M point moving average filter. The frequency, f, runs between 0 and 0.5. For $f = 0$, use: $H[f] = 1$

$$H[f] = \frac{\sin(\pi f M)}{M \sin(\pi f)}$$

The roll-off is very slow and the stopband attenuation is ghastly. Clearly, the moving average filter cannot separate one band of frequencies from another. Remember, good performance in the time domain results in poor performance in the frequency domain, and vice versa. In short, the moving average is an exceptionally good *smoothing filter* (the action in the time domain), but an exceptionally bad *low-pass filter* (the action in the frequency domain).

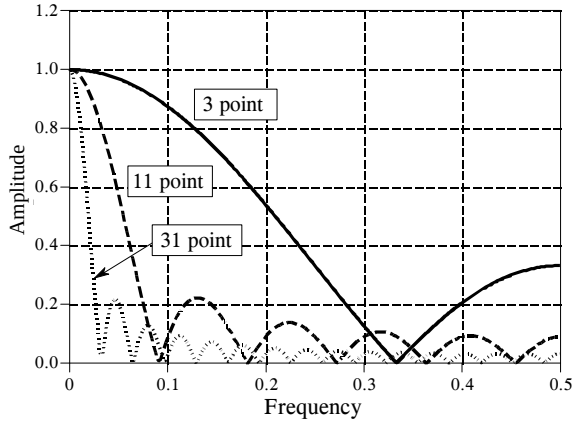

FIGURE 15-2
Frequency response of the moving average filter. The moving average is a very poor low-pass filter, due to its slow roll-off and poor stopband attenuation. These curves are generated by Eq. 15-2.

Relatives of the Moving Average Filter

In a perfect world, filter designers would only have to deal with time domain *or* frequency domain encoded information, but never a mixture of the two in the same signal. Unfortunately, there are some applications where both domains are simultaneously important. For instance, television signals fall into this nasty category. Video information is encoded in the time domain, that is, the shape of the waveform corresponds to the patterns of brightness in the image. However, during transmission the video signal is treated according to its frequency composition, such as its total bandwidth, how the carrier waves for sound & color are added, elimination & restoration of the DC component, etc. As another example, electromagnetic interference is best understood in the frequency domain, even if